Novozymes Enzyme Stability Prediction

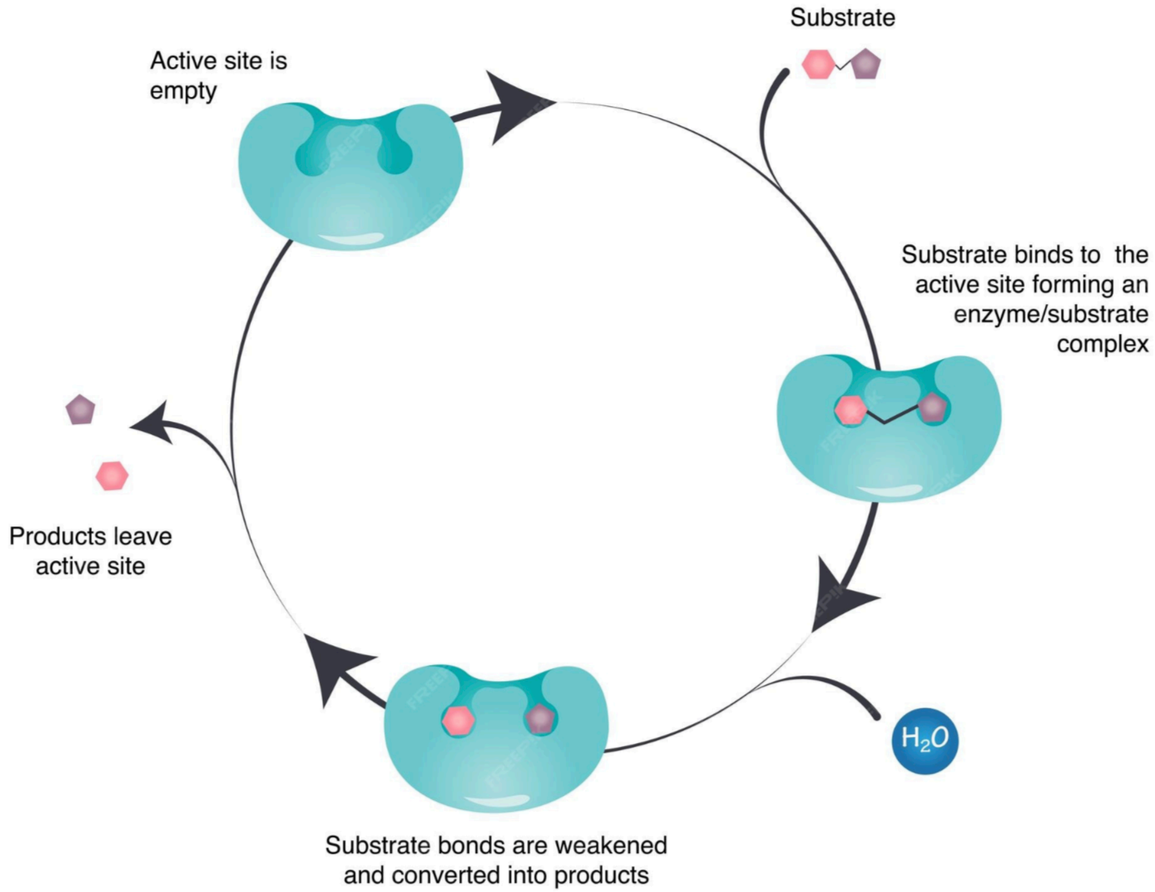
Kaggle Machine Learning Competition: 24th Silver Medal Solution; Joint work with Haoyuan Liu (Dynavax Technologies, UCLA)

Presenter: Chuanwan (Christine) Wu

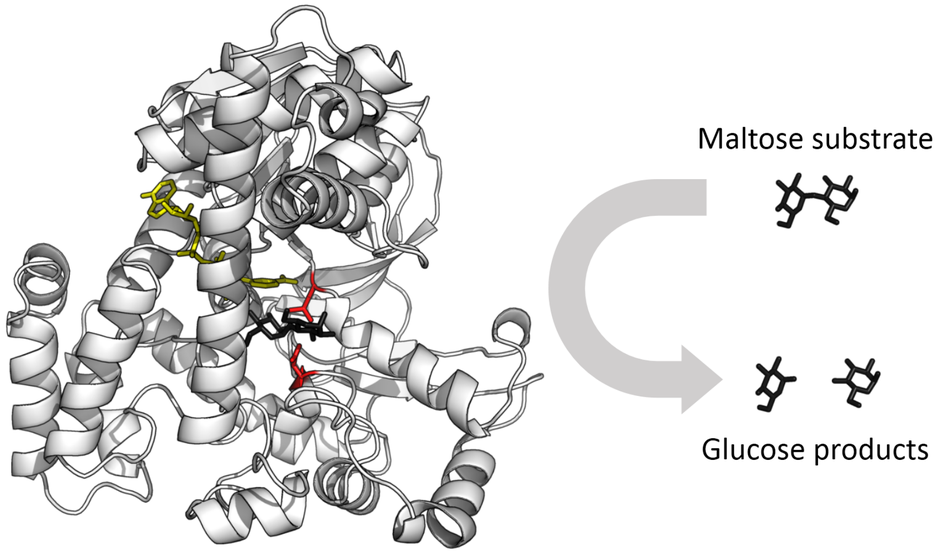
Introduction

Enzymes & their functionality

Enzymes are proteins that act as biological catalysts, speeding up chemical reactions in living organisms without being consumed in the process.



The catalytic cycle of an enzyme.

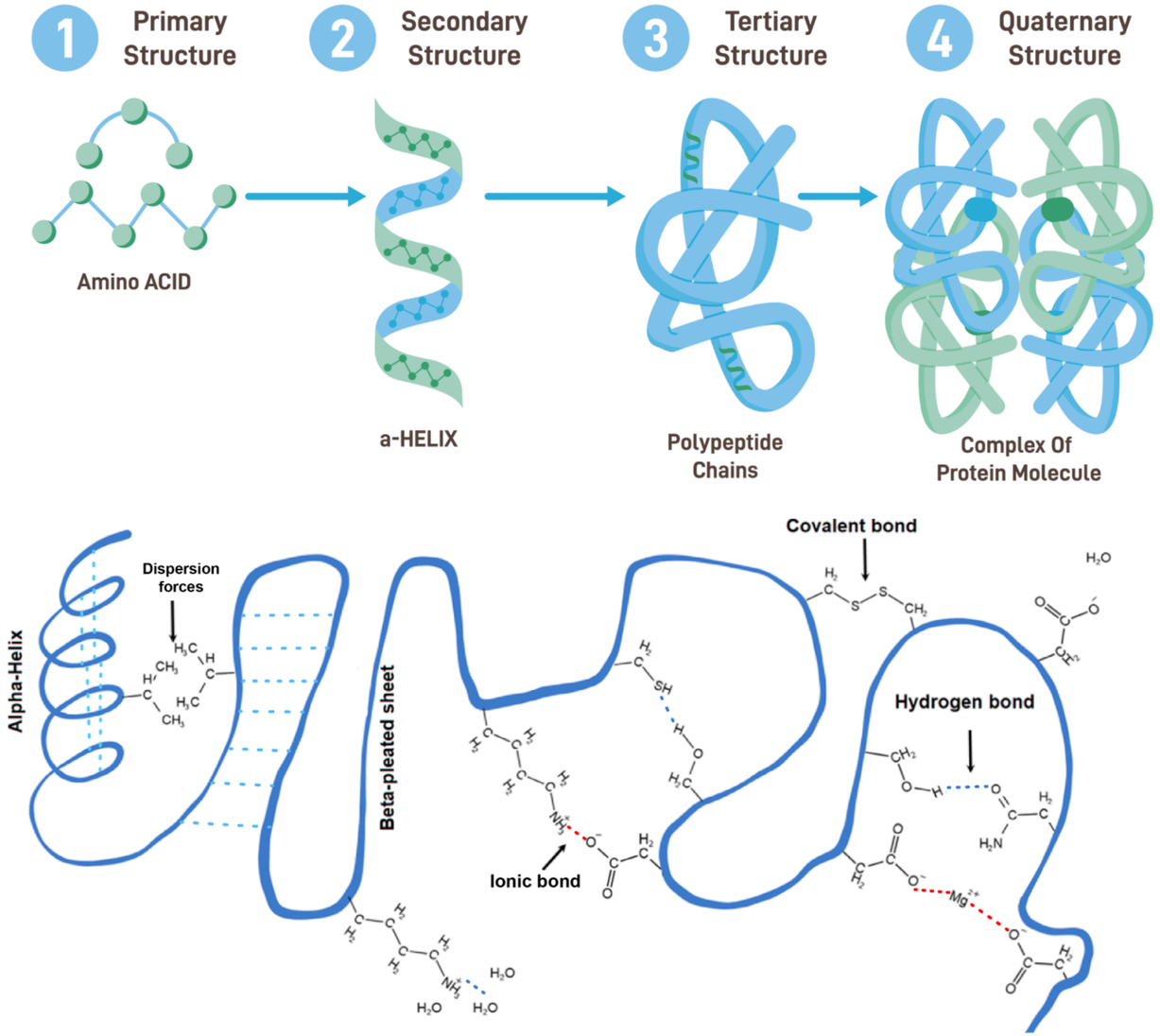


The enzyme *glucosidase* converts the sugar maltose into two glucose sugars. Active site residues in red, maltose substrate in black, and NAD cofactor in yellow.

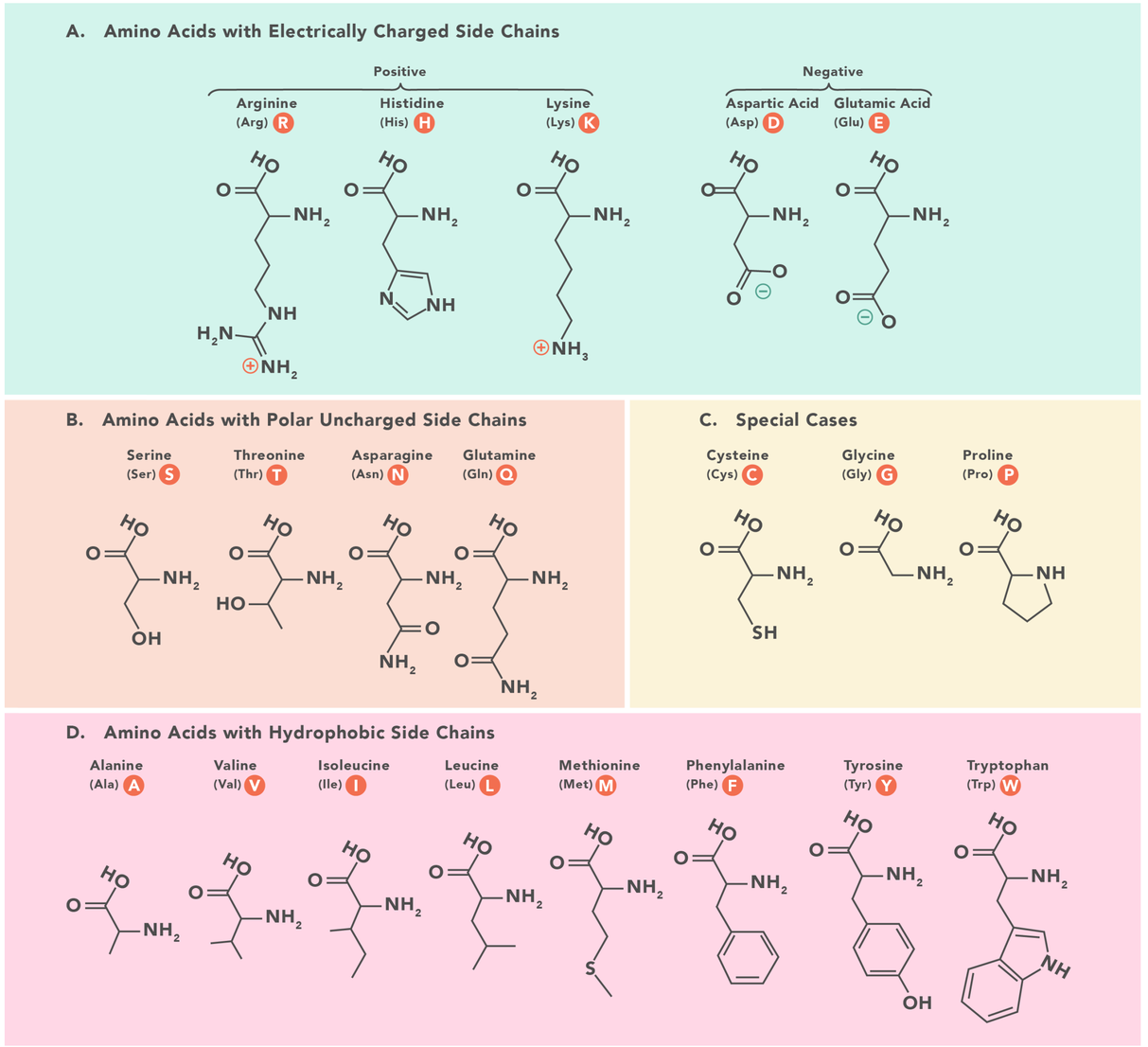
Wikipedia contributors. (n.d.). Enzyme. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Enzyme*

Freepik. (n.d.). Vector illustration of a diagram of an enzyme catalytic cycle template [Image]. Freepik. Retrieved May 23, 2024, from *https://www.freepik.com/premium-vector/vector-illustration-diagram-enzyme-catalytic-cycle-template_30160592.htm*

Enzymes as proteins *



Protein structural levels (top); molecular interactions on amino acids (bottom).



Amino Acid chart.

Monash University. (n.d.). Protein structure. Student Academic Success. Retrieved May 23, 2024, from *https://www.monash.edu/student-academic-success/chemistry/medicinal-chemistry/protein-structure*

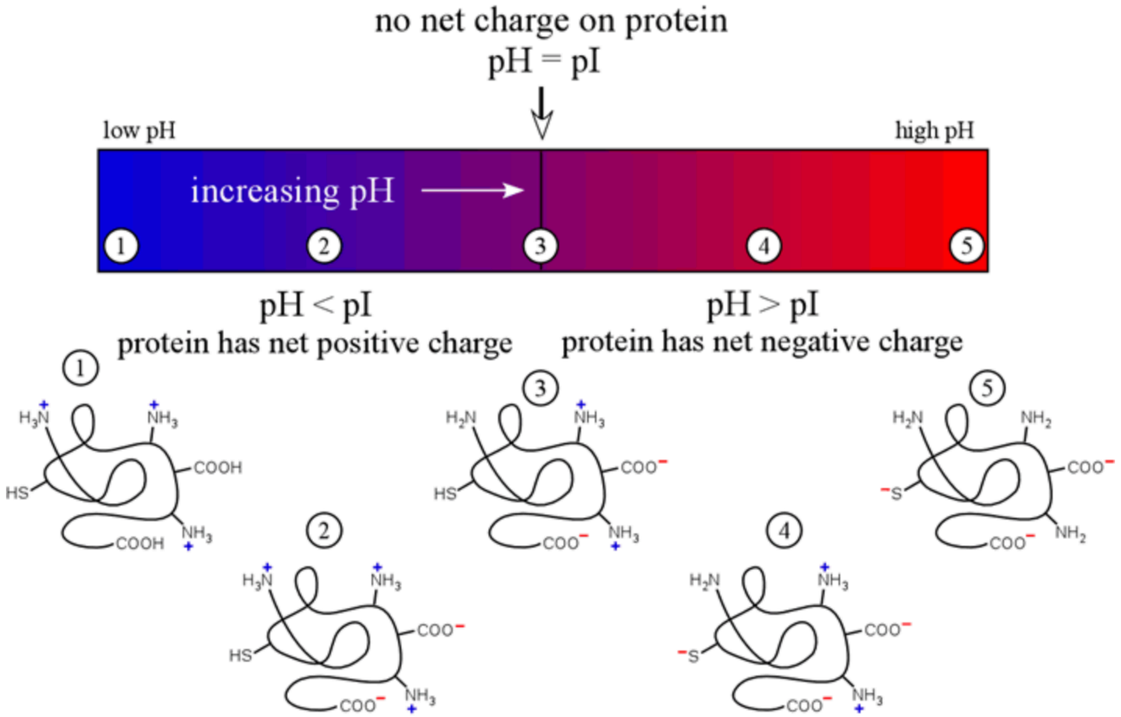
Steward, K., PhD. (2023, December 18). *Essential Amino Acids: Chart, Abbreviations and structure*. Applied Sciences From Technology Networks. *https://www.technologynetworks.com/applied-sciences/articles/essential-amino-acids-chart-abbreviations-and-structure-324357*

Proteins, pH & thermostability **

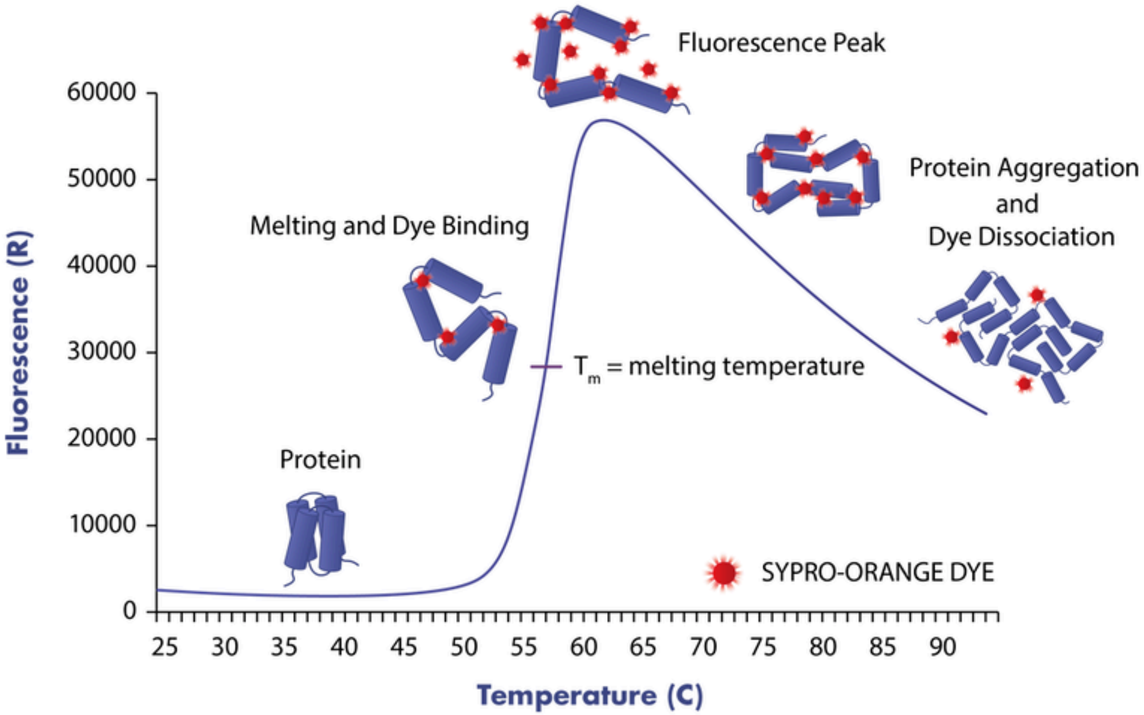
Cronk, J. D. (n.d.). Protein methods. Gonzaga University. Retrieved May 23, 2024, from *http://guweb2.gonzaga.edu/faculty/cronk/CHEM245pub/protein_methods.html*

Reaction Biology. (n.d.). Thermal Shift Assay (TSA). Retrieved May 23, 2024, from *https://www.reactionbiology.com/services/biophysical-assays/thermal-shift-assay-tsa*

The pH of a solution and temperature can dramatically affect a protein's stability, primarily by altering its state, which can lead to unfolding and loss of functionality.



How the charge on a protein would vary with the "ambient" pH of the medium of which it is a component.



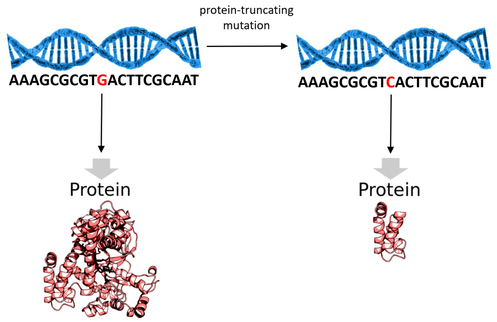
How the temperature can affect a protein's stability.

Proteins & mutations ***

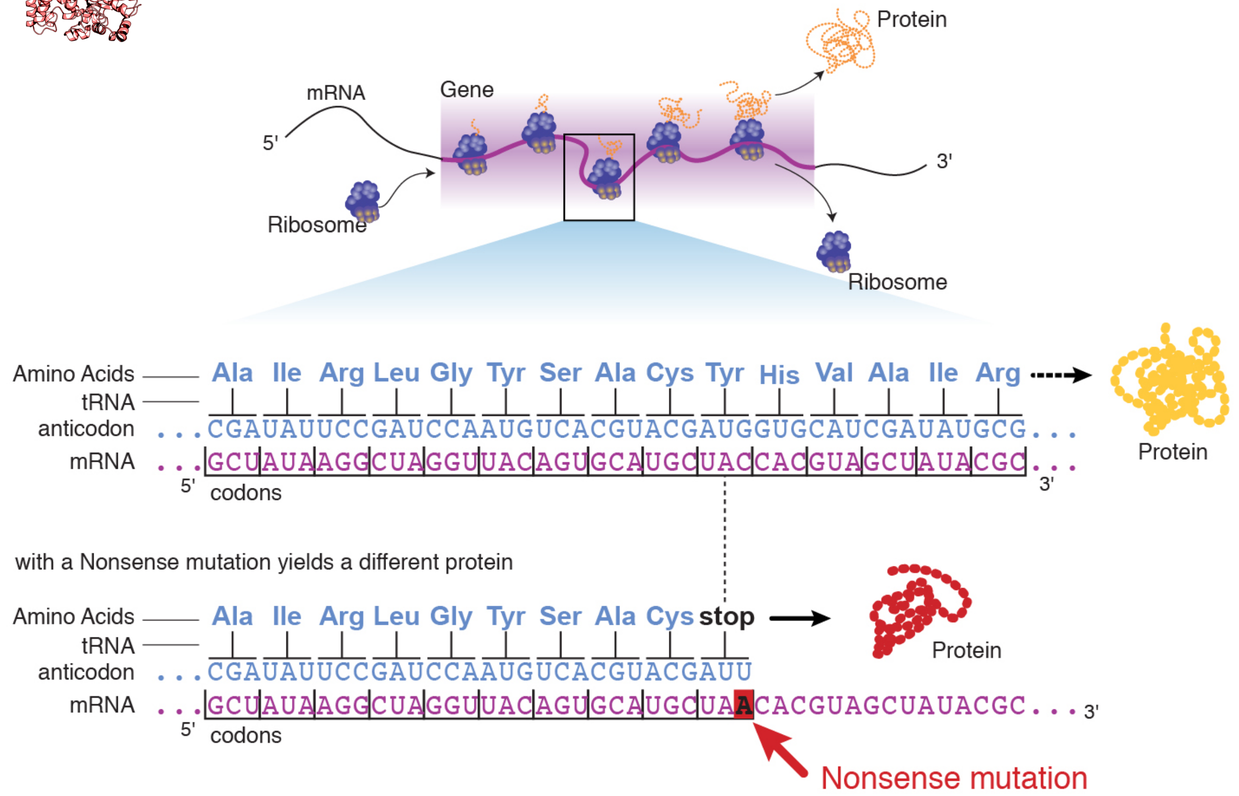
The Dish on Science. (n.d.). [Article Title]. Stanford University. Retrieved May 23, 2024, from *https://thedishonscience.stanford.edu/articles/ekg280kcyxt6r9dwmszsvgr1cxlwpj*

AR Inspired Pencil. (n.d.). DNA mutation deletion [Image]. Retrieved May 23, 2024, from *https://ar.inspiredpencil.com/pictures-2023/dna-mutation-deletion*

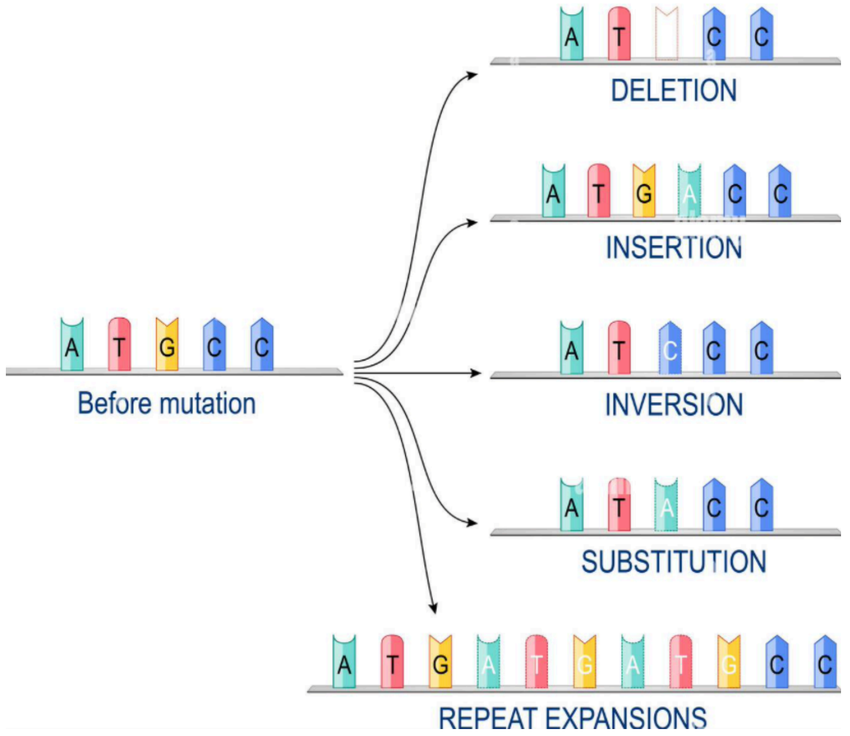
Oncology Nursing Society. (n.d.). Variant. ONS.org. Retrieved May 23, 2024, from *https://onf.ons.org/variant*



A mutation first occurs in the DNA sequence of the gene that codes for the protein. It then passes through mRNA in transcription, and mRNA to protein in translation.



Mutations allow us to redesign natural enzymes to withstand harsh industrial conditions.

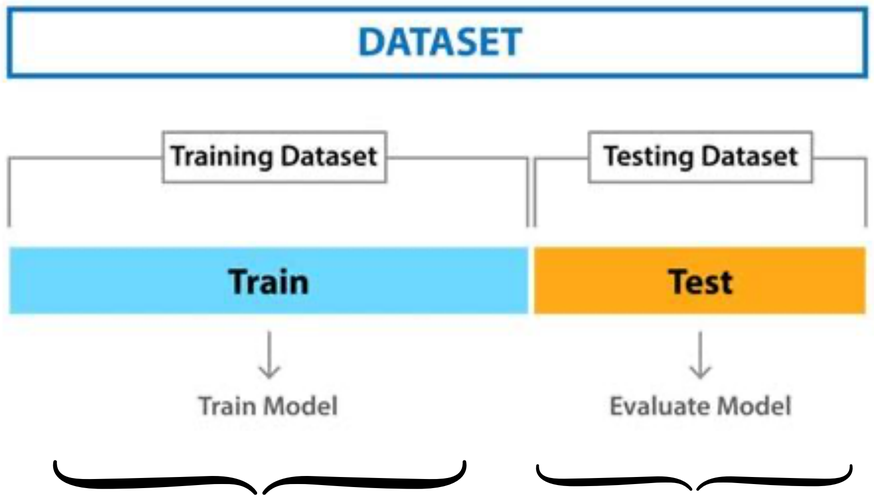


Different types of mutations.

Objective

Training & testing datasets

VProExpert. (n.d.). Machine learning model validation. Retrieved May 23, 2024, from *https://www.vproexpert.com/machine-learning-model-validation/*



i. Protein Sequence (**Enzymes as proteins** *)
ii. pH (**Protein, pH & thermostability** **)
iii. Melting point tm (**Protein, pH & thermostability** **)
iv. Sequence id
v. Data source

i. Single-mutation variant of an enzyme (**Protein & mutations** ***)

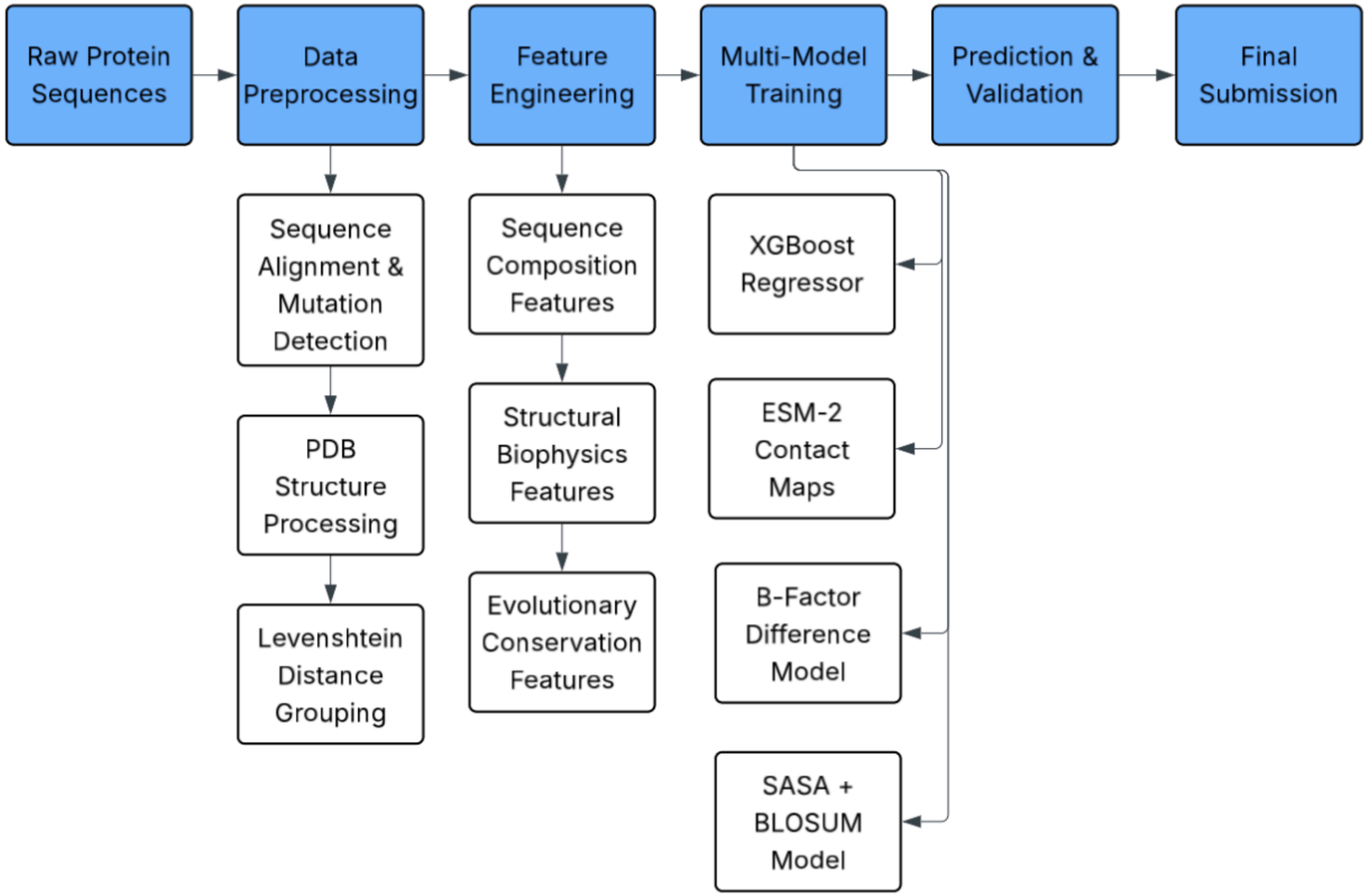
Objective:

Predict how a single change (mutation or deletion) to a protein's sequence affects its ability to withstand heat.

Methods

Methodology Overview

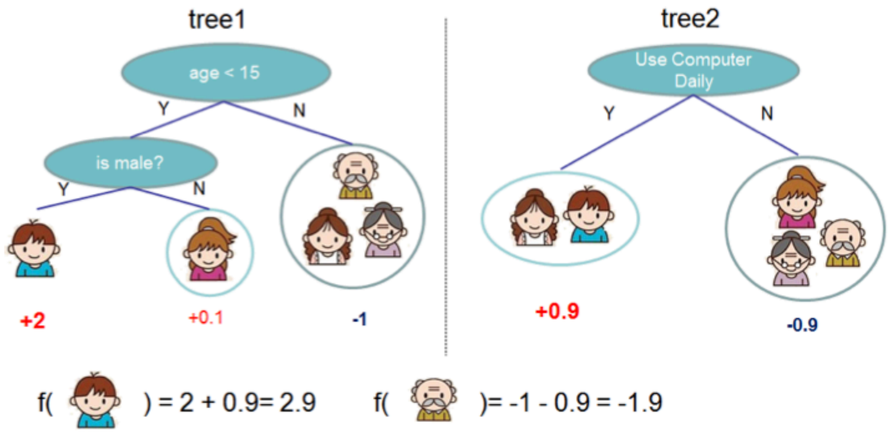
Process flow chart:



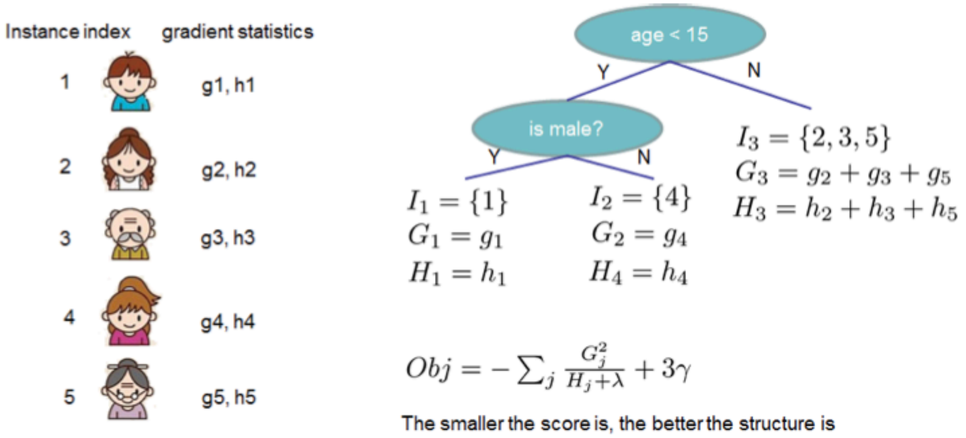
** Note: The PDB files were taken from the kaggle community, and the code details can be checked at the Github page.*

I. Extreme Gradient Boosting (XGBoost) Model

*Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794). ACM. *https://doi.org/10.1145/2939672.2939785*



Tree Ensemble Model. The final prediction for a given example is the sum of predictions from each tree.



Structure Score Calculation. We only need to sum up the gradient and second order gradient statistics on each leaf, then apply the scoring formula to get the quality score.

We minimize the objective function by sequentially adding trees that optimally reduce loss while controlling complexity.

*Derivations (for more details, check *):*

Step 1. Main objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \qquad (*)$$

where

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2$$

Step 2. At iteration t (remove the constant terms):

$$\tilde{\mathcal{L}}^{(t)} = \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \qquad (1)$$

Step 3. Define $I_j = \{i | q(\mathbf{x}_i) = j\}$ as the instance set of leaf j. We can rewrite Eq (1) by expanding Ω as follows:

$$\begin{aligned} \tilde{\mathcal{L}}^{(t)} &= \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \\ &= \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \end{aligned} \qquad (2)$$

Step 4. For a fixed structure $q(\mathbf{x})$, we can compute the optimal weight w_j^* of leaf j by:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \qquad (3)$$

II. ESM-2

Hsu, C., Nisonoff, H., Fannjiang, C., & Listgarten, J. (2020). Learning inverse folding from millions of predicted structures. BioRxiv. *https://doi.org/10.1101/2020.12.15.422761*

Cheng, J., et al. (2022). ESM Metagenomic Atlas: The first view of the 'dark matter' of the protein universe. BioRxiv. *https://doi.org/10.1101/2022.07.20.500902*

Ramith H. (n.d.). *ESM-2: Evolutionary-scale prediction of atomic-level protein structure with a language model*. Ramith.fyi. Retrieved May 23, 2024, from *https://ramith.fyi/esm-2-evolutionary-scale-prediction-of-atomic-level-protein-structure-with-a-language-model/*

ESM-2 learns protein sequences through masked language modeling, and we discover its attention patterns using logistic regression (*).

(*)

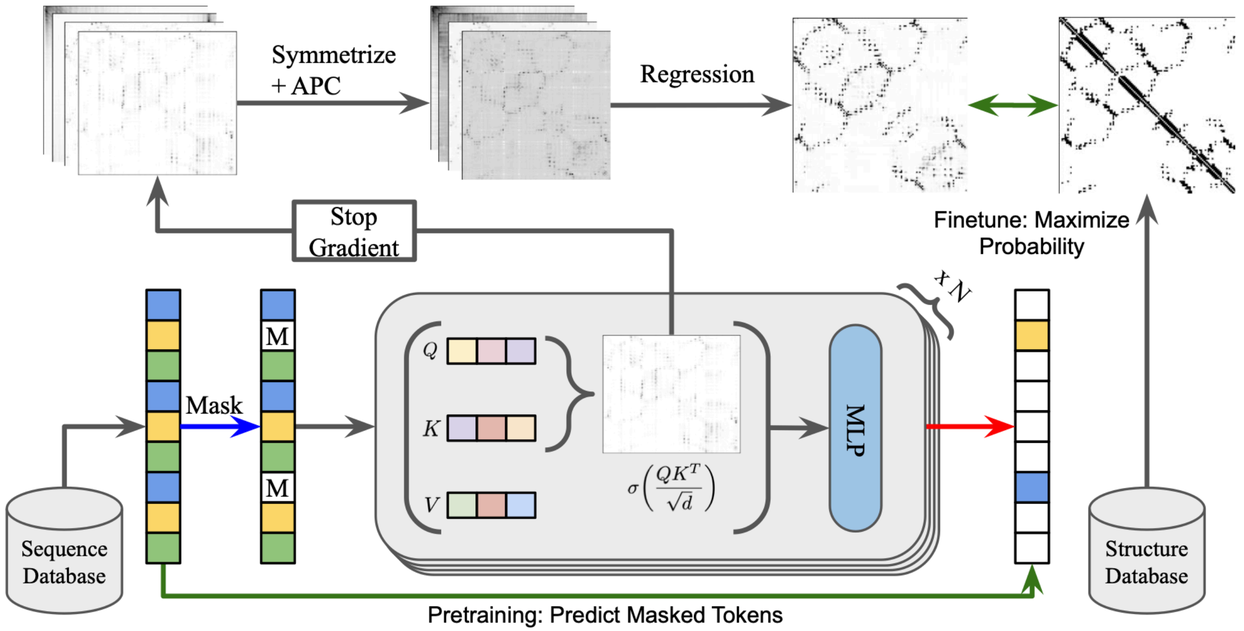
$$p(c_{ij}) = \left(1 + \exp\left(-\beta_0 - \sum_{l=1}^{L}\sum_{k=1}^{K}\beta_{kl}a_{ij}^{kl}\right)\right)^{-1}$$

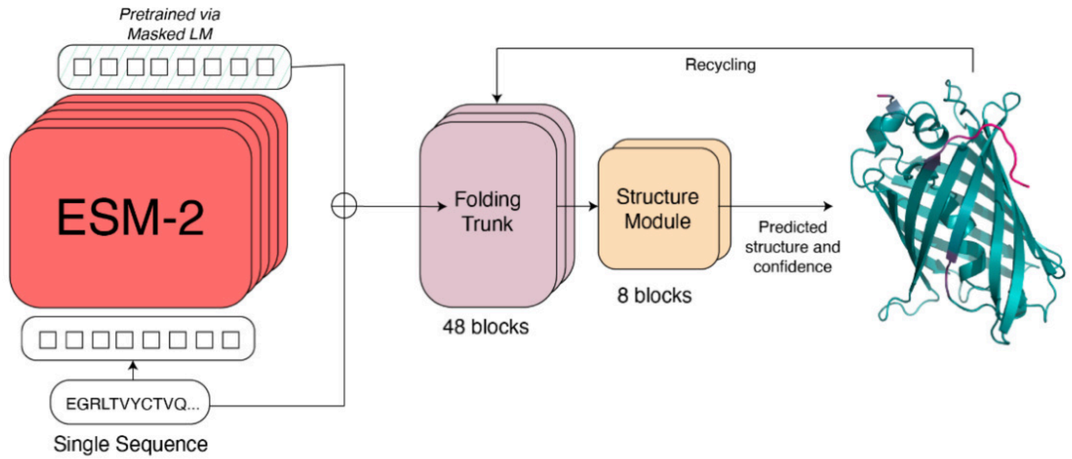
value of the attention map at i,j

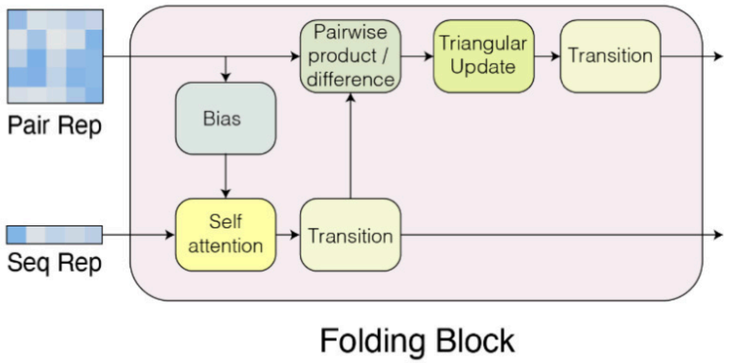
RV {True , False} , which is true if amino acids i, j are in contact

K = # of attention heads

L = # of layers

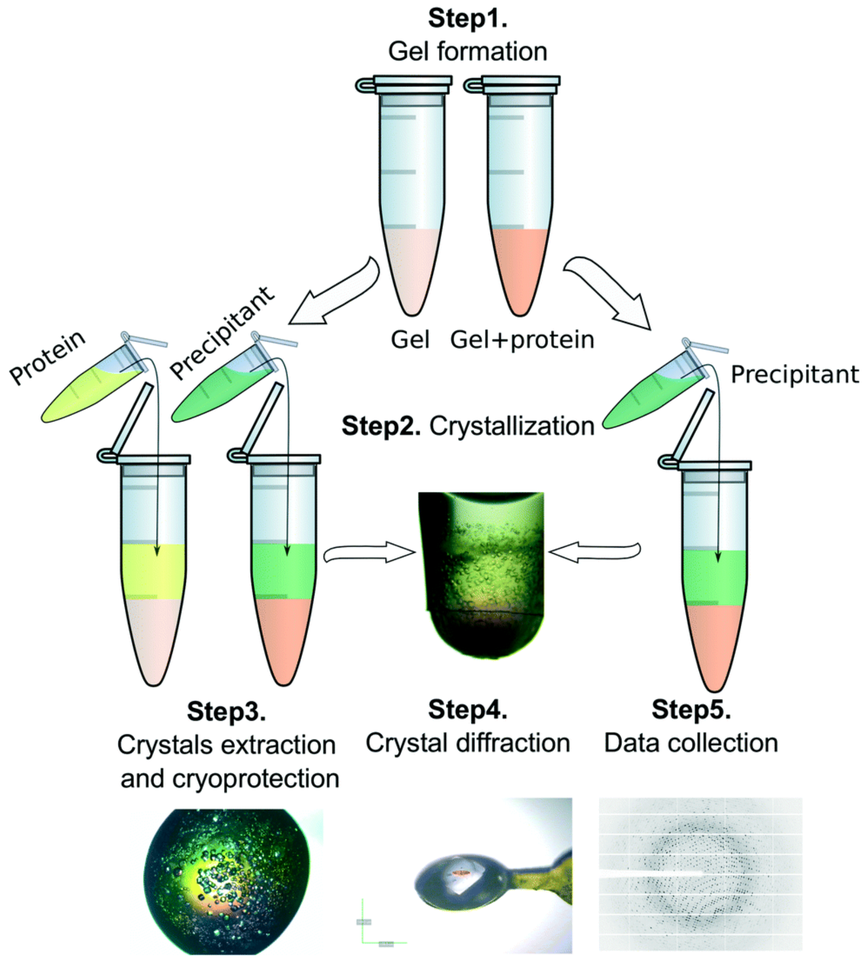




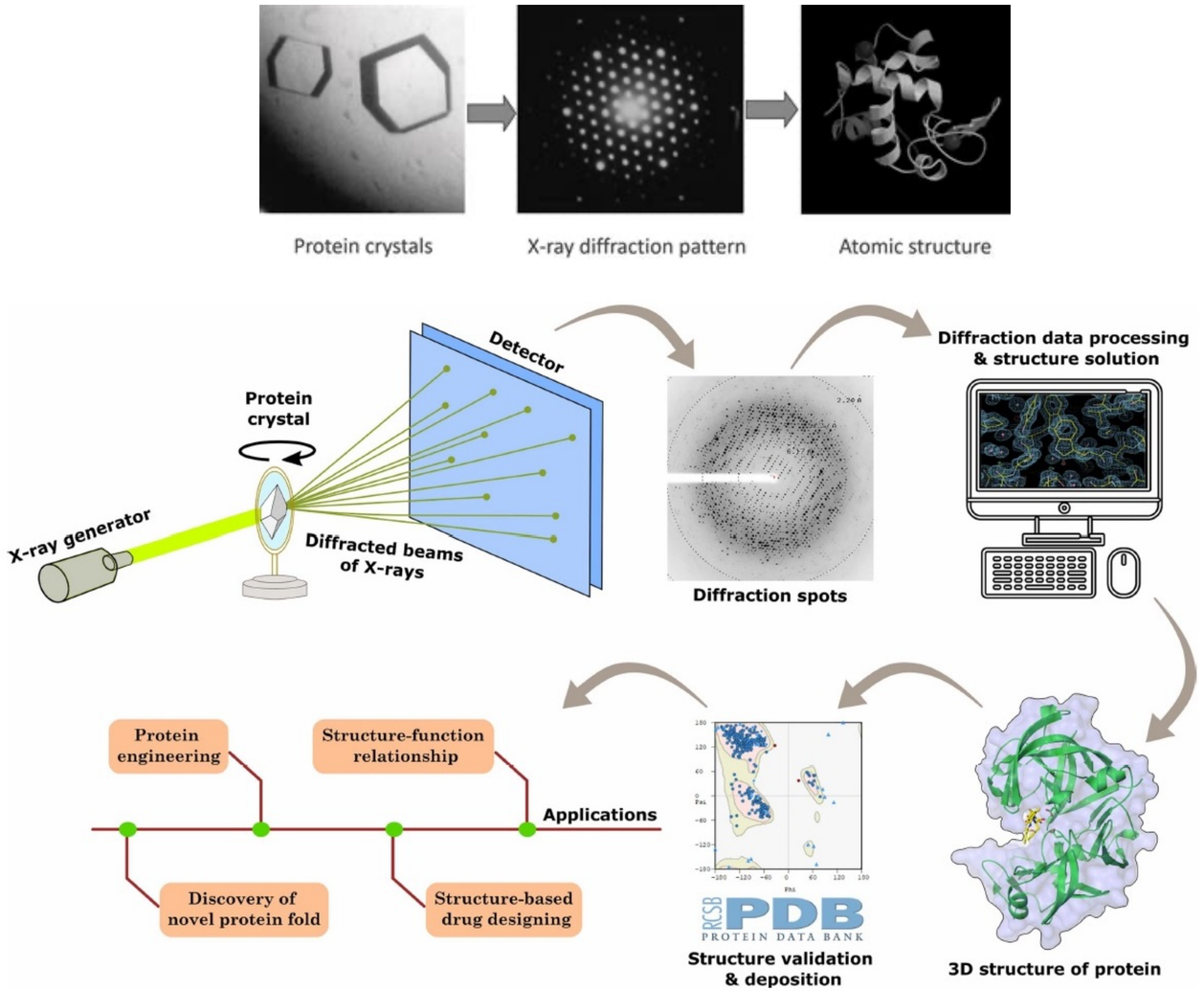


III. Pure Chem Model (B-factor & SASA)

What are B-factor, SASA, and how they're related to theromokinetics?
First of all, proteins (e.g., enzymes) can be crystallized.



Standard workflow to prepare protein X-ray crystallography.



X-ray crystallography pipeline and its applications.

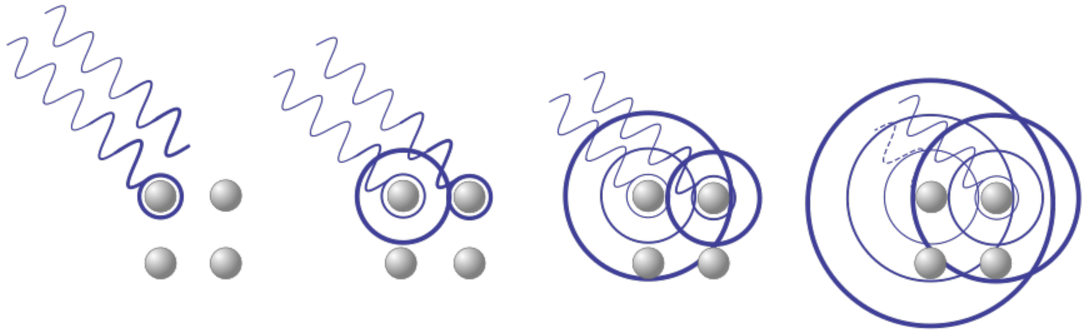
Bhaumik, P. (n.d.). Prasanta Bhaumik's Homepage. Department of Biosciences and Bioengineering, IIT Bombay. Retrieved May 23, 2024, from *https://www.bio.iitb.ac.in/~pbhaumik/*

AR Inspired Pencil. (n.d.). Protein crystallization process [Image]. Retrieved May 23, 2024, from *https://ar.inspiredpencil.com/pictures-2023/protein-crystallization-process*

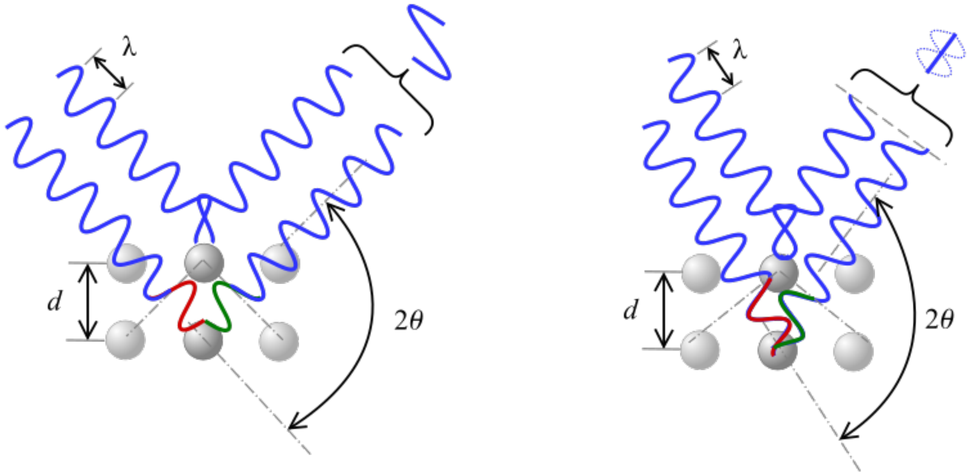
III. i. Bragg's Law & Crystallography

Wikipedia contributors. (n.d.). Bragg's law. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Bragg%27s_law*

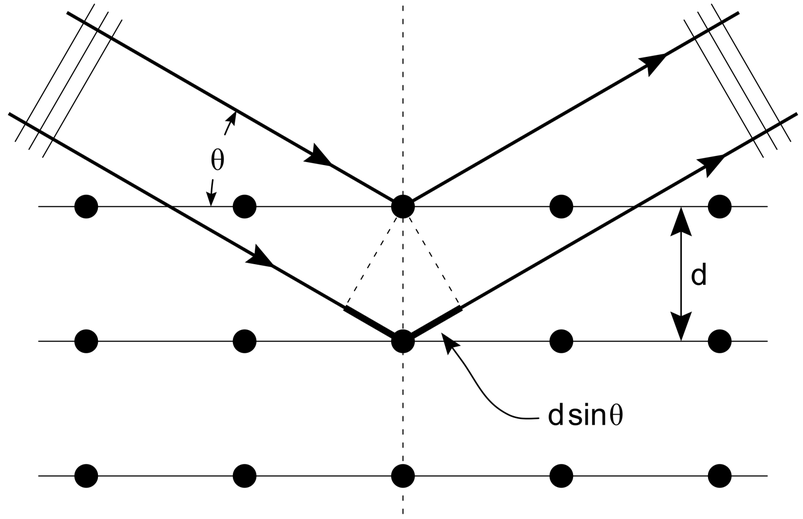
Crystals, at certain specific wavelengths and incident angles, produced intense peaks of reflected radiation.



X-rays interact with the atoms in a crystal.



According to the 2θ deviation, the phase shift causes constructive (left figure) or destructive (right figure) interferences.



$$n\lambda = 2d\sin\theta$$ * (Bragg's Law)

Constructive interference (Bragg diffraction) occurs when X-rays scattering off crystal planes satisfy Bragg's Law, where the path difference is an integer multiple of the wavelength.

* where λ is the wavelength, n is the diffraction order, θ is the glancing angle, and d is the glancing constant.

III. i. From Bragg's Law to B-factors

Wikipedia contributors. (n.d.). Debye–Waller factor. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Debye%E2%80%93Waller_factor*

B-factor is quantified from how much atomic "shaking" reduces diffraction intensities via the Debye-Waller factor.

Derivations:

Step 1. The Debye-Waller factor describes how atomic thermal motion reduces scattering:

$$f_{DW} = \langle \exp(i\mathbf{s}\cdot\mathbf{u})\rangle \approx 1 + i\mathbf{s}\cdot\langle\mathbf{u}\rangle - \frac{1}{2}\langle(\mathbf{s}\cdot\mathbf{u})^2\rangle + \cdots \tag{1}$$

where $\mathbf{s}$ is the scattering vector ($|\mathbf{s}| = 2\sin\theta/\lambda$); $\mathbf{u}$ is the atomic displacement from mean position; $\langle\cdot\rangle$ is the thermal average.

Step 2. For isotropic thermal motion:

i. $\langle\mathbf{u}\rangle = 0$ (no net displacement)

ii. $\langle u_x^2\rangle = \langle u_y^2\rangle = \langle u_z^2\rangle = \langle u^2\rangle/3$

Then:

$$\langle(\mathbf{s}\cdot\mathbf{u})^2\rangle = s^2\langle u^2\rangle\langle\cos^2\phi\rangle = s^2\langle u^2\rangle\cdot\frac{1}{3} \tag{2}$$

Step 3. We then get a Gaussian distribution:

$$\langle \exp(i\mathbf{s}\cdot\mathbf{u})\rangle = \exp\left(-\frac{1}{6}s^2\langle u^2\rangle\right) \tag{3}$$

Step 4. With s inserted, we have:

$$f_{DW} = \exp\left(-\frac{1}{6}\cdot\frac{4\sin^2\theta}{\lambda^2}\cdot\langle u^2\rangle\right) = \exp\left(-\frac{2}{3}\cdot\frac{\sin^2\theta}{\lambda^2}\cdot\langle u^2\rangle\right) \tag{4}$$

Step 5. By convention, we define the *B-factor* as:

$$\boxed{B = 8\pi^2\langle u^2\rangle} \tag{*}$$

B-factor

Substituting $\langle u^2\rangle = B/8\pi^2$:

$$f_{DW} = \exp\left(-\frac{2}{3}\cdot\frac{\sin^2\theta}{\lambda^2}\cdot\frac{B}{8\pi^2}\right) = \exp\left(-\frac{B}{12\pi^2}\cdot\frac{\sin^2\theta}{\lambda^2}\right) \tag{5}$$

Step 6. The standard form used in crystallography is:

$$\boxed{f_{DW} = \exp\left[-B\cdot\frac{\sin^2\theta}{\lambda^2}\right]} \tag{**}$$

Debye-Waller factor

III. ii. Solvent Accessible Surface Area (SASA)

The accessible surface area (ASA) or solvent-accessible surface area (SASA) is the surface area of a biomolecule that is accessible to a solvent.

*SASA Definitions (for more details, check *):*

i. Lee & Richards method (Rolling Probe):

$$\text{SASA} = \text{Area traced by center of solvent sphere}$$

ii. Shrake & Rupley algorithm:

$$\text{SASA} \approx \sum(\text{accessible surface points}) \times (\text{area per point})$$

iii. ΔSASA upon binding:

$$\Delta\text{SASA} = \text{SASA}_{\text{complex}} - (\text{SASA}_{\text{protein}} + \text{SASA}_{\text{ligand}})$$

where negative values indicate burial upon binding.

iv. Hydrophobic effect & folding:

$$\Delta G_{\text{folding}} \propto \Delta\text{SASA}_{\text{hydrophobic}} \times \gamma$$

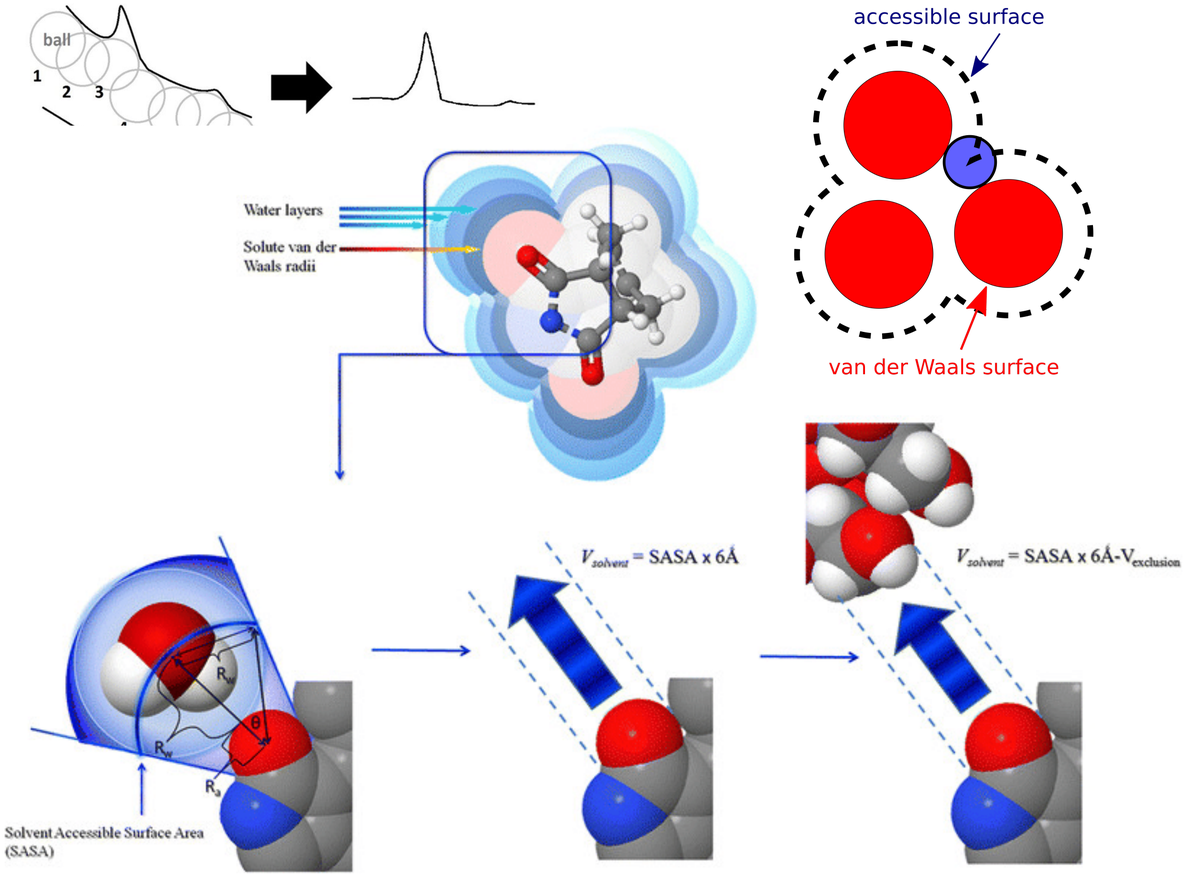
where $\gamma \approx 20 - 25\ \text{cal} \cdot \text{mol}^{-1} \cdot \text{Å}^{-2}$ (hydrophobic energy density).

v. Transfer free energy:

$$\Delta G_{\text{transfer}} = \sum \sigma_i \cdot \text{SASA}_i$$

where σ_i = atomic solvation parameters.

Schematic of the rolling ball algorithm & molecular motions.



* ScienceDirect. (n.d.). Accessible surface area. In ScienceDirect Topics. Elsevier. Retrieved May 23, 2024, from *https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/accessible-surface-area*

Niskanen, J., & Laine, S. (2017). A novel method to remove the background from X-ray diffraction signal. Journal of Applied Crystallography, 50(6), 1860–1863. *https://doi.org/10.1107/S1600576717015234*

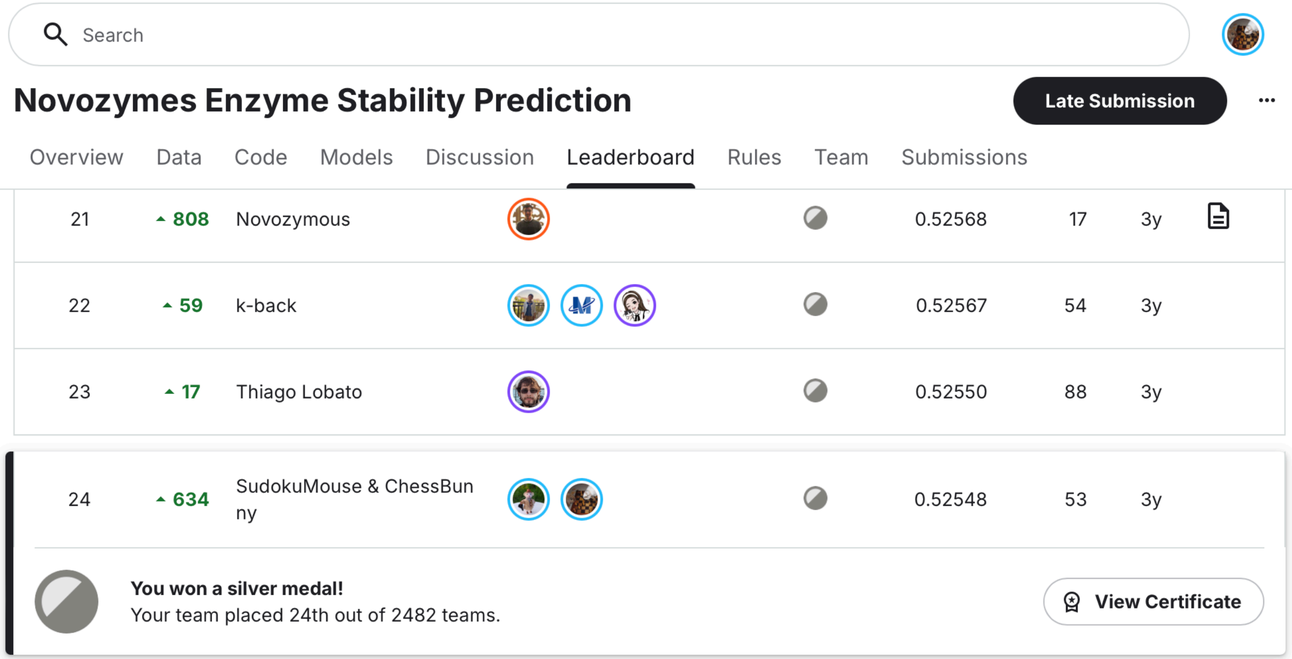
Liu, S., Cai, C., Tsai, H., Cao, L., Tan, Y., & Gervasio, F. L. (2017). On the fly estimation of host–guest binding free energies using the movable type method: participation in the SAMPL5 blind challenge. *Journal of Computer-Aided Molecular Design, 31*(1), 47–60. *https://doi.org/10.1007/s10822-016-9930-3*

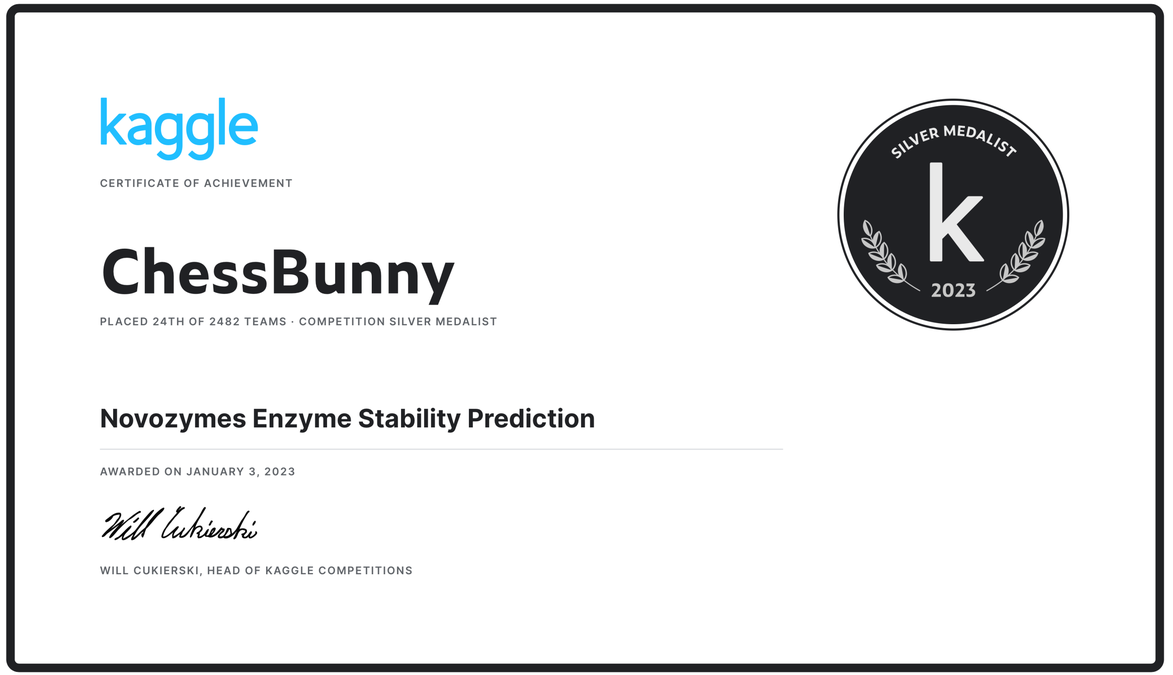
Wikipedia contributors. (n.d.). Accessible surface area. In Wikipedia. Retrieved May 23, 2024, from *https://en.wikipedia.org/wiki/Accessible_surface_area*

Results

Results

We won a silver medal!

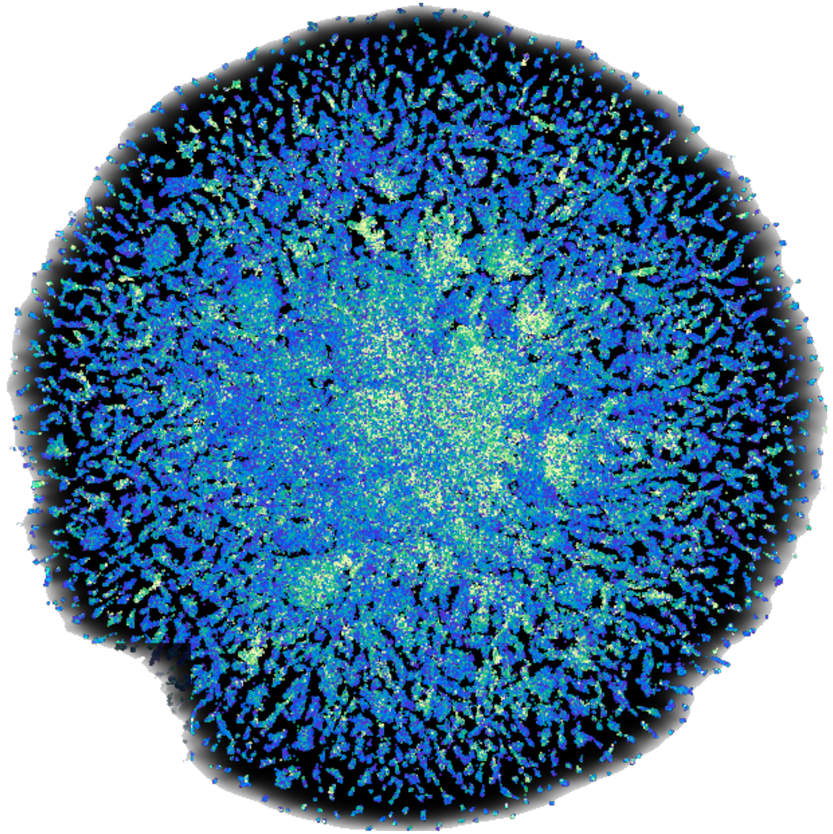




(Yes, I call myself chessbunny.)

Conclusions

Conclusions & Recommendations



Exploring 1 million out of 772M proteins (ESM).

AI is revolutionizing our ability to predict protein stability, opening up incredible new possibilities for discovery. By embracing its power while being mindful of its current limitations, we can guide it to even greater breakthroughs.